

TECHEX HACKATHON 2026 · TRACK 1: AGENT SECURITY · VEEA SPONSOR · MAY 19 DEADLINE

SOUF AI

Sub-millisecond LLM governance · Open source · Self-hosted

F1 = 1.000 · 231 PROMPTS

5 benchmarks · 188 TP · 43 TN · 0 FP · 0 FN · Wilson 95% CI DENY [0.980, 1.000]

0.051 ms

P50 DPI LATENCY · 1000× FASTER THAN LAKERA

17,553 req/s

SINGLE-CORE THROUGHPUT · OFFLINE-CAPABLE

337 rules · 16 sets

HIPAA · PCI-DSS · OWASP LLM01-10

Track 1: Agent Security · Veeva Award Stack · NVIDIA DGX Spark Prize · MIT License

Sardor Razikov · Independent ML Engineer · Top 1.9% Kaggle SPR · ECB v2 Zenodo Author · ORCID: 0009-0007-0731-4247 · Tashkent, Uzbekistan

EXTENDS VEEA'S MIT LOBSTER TRAP · COMMERCIAL ALTERNATIVE TO LAKERA / NEMO / PROMPT GUARD

Lobster Trap is the floor. SOUF AI is the ceiling — built solo from Tashkent for the TechEx Veeva hackathon.

github.com/SRKRZ23/souf-ai · python3 scripts/run_all_benchmarks.py · 3.3 seconds end-to-end

Enterprise LLM applications leak data. The clock is ticking.

Existing proxies block 39.6% of attacks. EU enforcement starts **August 2, 2026.**

EU AI ACT · ARTICLE 9 · AUGUST 2, 2026

"Providers of high-risk AI systems shall establish, implement, document and maintain a risk management system."

High-risk AI faces fines up to €30M or 6% of global turnover.

€30M EXPOSURE

eur-lex.europa.eu · EU AI Act Art. 9(1)

OWASP LLM TOP 10 · 2025 EDITION · owasp.org

LLM01: Prompt Injection rated #1 critical risk. "Attackers manipulate LLMs through crafted inputs."

Published standard — every enterprise deployment must address all 10 categories.

#1 CRITICAL RISK

owasp.org/www-project-top-10-for-large-language-model-applications

TECHCRUNCH · LAKERA SERIES A · 2024

Lakera raises \$20M to guard against prompt injection. No encoding attack defense. No HIPAA pack. Proprietary.

Rebuff.ai: abandoned. Prompt Security: proprietary. LLM Guard: no compliance packs.

\$20M · GAPS REMAIN

techcrunch.com · Crunchbase · public docs 2024

LOBSTER TRAP BASELINE (Veea, MIT)

19 / 48 blocked · F1 = 0.567 · 39.6% block rate

SOUF AI v0.5f (this submission)

48 / 48 blocked · F1 = 1.000 · +60.4 percentage points

HIPAA BREACH COST

\$4.35M avg

EU AI ACT MAX FINE

€30M or 6%

AI SECURITY MARKET

\$4.1B · 2025

OPEN-SOURCE SOLUTIONS

ZERO. Until now.

Lobster Trap baseline (Veea, MIT): 19/48 = 39.6% block rate, F1=0.567 · SOUF AI v0.5f: 48/48 = 100%, F1=1.000

Regulators defined the requirement.

SOUF AI is the open-source proof they're asking for.

EU AI ACT · ARTICLE 9 · FULL TEXT

"Providers of high-risk AI systems shall establish, implement, document and maintain a risk management system that is in continuous iterative process throughout the entire lifecycle."

European Parliament · EU AI Act Art. 9(1) · Enforcement: August 2, 2026

NIST AI RMF · GOVERN 1.1 · PUBLISHED 2023

"Policies, processes, procedures and practices across the organization related to the mapping, measuring, and managing of AI risks are in place, transparent, and implemented effectively."

NIST AI Risk Management Framework · US National Institute of Standards and Technology

SOUF AI DELIVERS EXACTLY THIS:

- ✓ 231 prompts tested — F1 = 1.000
- ✓ Wilson 95% CI DENY: [0.980, 1.000]
- ✓ HIPAA §164.502 compliant rule pack
- ✓ PCI-DSS Req 3.3 compliant rule pack
- ✓ Ed25519-signed audit trail per request
- ✓ Upstream PR submitted to veeainc/lobstertrap
- ✓ MIT licensed — zero vendor lock-in
- ✓ NFKC normalize + 54 confusable codepoints
- ✓ 14 PatternSets · 254+ regex patterns

github.com/SRKZR23/souf-ai · [scripts/run_all_benchmarks.py](#)

THEY SAID IT. WE BUILT THE PROOF.

Every claim above is empirically verified · Source: [python scripts/run_all_benchmarks.py](#) → 231 prompts · ALL PASS

DROP-IN PROXY · ZERO LLM MODIFICATION · P99 = 0.111ms

Deep Packet Inspection — Sits Between App and LLM



No model changes. No API changes. No data leaves your infra. Drop-in proxy.

MEDIAN LATENCY

0.051ms

P50 · Apple M2

P99 LATENCY

0.111ms

99th percentile · all 231 prompts

THROUGHPUT

17,553

requests / second

PATTERNSET COVERAGE

OWASP LLM01:	Prompt injection, jailbreak, role-play escapes, system prompt extraction · 48/48 PASS
HIPAA §164.502:	PHI disclosure patterns · SSN, MRN, DOB, diagnosis codes · 20/20 PASS
PCI-DSS Req 3.3:	PAN disclosure, CVV exfiltration, card data leakage · 20/20 PASS
ENCODING ATTACKS:	Token-split, fullwidth unicode, Cyrillic homoglyph, ROT-13 bypass · 23/23 PASS
OOD HELD-OUT:	100 never-seen adversarial prompts, zero in training data · 100/100 PASS

5 INDEPENDENT SUITES · 231 TOTAL PROMPTS · ZERO EXCEPTIONS · ALL F1=1.000

It works. Here are the numbers.

1.000

F1 across all 5 suites

BENCHMARK SUITE	PROMPTS	TRUE POS	FALSE POS	FALSE NEG	F1 SCORE	STATUS
In-distribution v0.5c	68	48	0	0	1.000	✓ PASS
OOD held-out	100	90	0	0	1.000	✓ PASS
HIPAA §164.502	20	16	0	0	1.000	✓ PASS
PCI-DSS Req 3.3	20	16	0	0	1.000	✓ PASS
Encoding Attacks #5	23	18	0	0	1.000	✓ PASS
TOTAL — ALL 5 SUITES	231	188	0	0	1.000	✓ ALL PASS

Wilson 95% CI — DENY: [0.980, 1.000] · ALLOW: [0.918, 1.000]Computed from n=231, TP=188, TN=43 — statistically significant at $p < 0.001$

FALSE POSITIVES

ZERO

FALSE NEGATIVES

ZERO

OOD GENERALIZATION

100/100

LOBSTER TRAP BASELINE

39.6% → 100% +60.4pp

MARKET LANDSCAPE · 2026 · ZERO FUNDED COMPETITORS PASS ALL 5 SUITES

Every competitor has a gap. SOUF AI has none.

Open-source. HIPAA. PCI. Encoding attacks.

All 5 suites. F1=1.000.

PRODUCT	OPEN-SOURCE	HIPAA	PCI-DSS	ENCODING ATTACKS	ALL 5 F1=1.000
Lakera Guard \$20M raised · Series A	✗ Proprietary	✗ No	✗ No	✗ No	✗ No
Rebuff.ai Abandoned	✓ MIT	✗ No	✗ No	✗ No	✗ No (abandoned)
Prompt Security Proprietary SaaS	✗ Proprietary	✗ No	✗ No	✗ No	✗ No
LLM Guard Open-source · no compliance	✓ Apache 2	✗ No	✗ No	✗ No	✗ No
SOUF AI v0.5f This submission · MIT · 231 prompts · ALL PASS	✓ MIT	✓ Yes	✓ Yes	✓ Yes	✓ F1=1.000

MARKET GAP · COMPETITOR ANALYSIS · 2026

"\$20M raised. Zero encoding attack defense.
Zero HIPAA pack. Proprietary. Per-seat fees."

SOUF AI is MIT-licensed and passes every suite they miss.

Crunchbase · public product docs as of 2026-05-13

THE VERDICT

Lobster Trap is the floor.
SOUF AI is the ceiling.

MIT licensed · Zero per-seat fees · Upstream PR submitted

SOUF AI is the security layer. The ecosystem uses it.

SOUF AI

THIS SUBMISSION

DPI rule packs · F1=1.000 · 231 prompts
14 PatternSets · 254+ regex patterns
Lobster Trap extension · MIT licensed
HIPAA §164.502 · PCI-DSS Req 3.3
Ed25519-signed audit trail per request
Upstream PR → [veeainc/lobstertrap](https://github.com/veeainc/lobstertrap)
P99 = 0.111ms · 17,553 req/s

DPI patterns

DPI per-call

FORGE

IBM Bob Hackathon · May 2026

IBM Bob codebase scanner
YAML policy generator
SOUF AI patterns embedded inside
Auto-generates SOUF AI rule packs
25/25 PASS · F1=1.000

github.com/SRKRZ23/forge-ai

ATLAS

AI Agent Olympics Hackathon · May 2026

Enterprise agent framework
SOUF AI DPI on every LLM call
Ed25519 audit trail per agent action
EU AI Act Art. 9 compliant by design
29/29 PASS · F1=1.000

github.com/SRKRZ23/atlas-agent

CITADEL

Gemma 4 Good Hackathon · May 2026

Open AI evaluation framework
13 evaluation layers (L0-L12)
Gemma 4 #3 on ECB v2 benchmark
SOUF AI patterns in L3 content safety
49/49 PASS · All levels green

github.com/SRKRZ23/citadel-eval

FORGE auto-generates SOUF AI policies for any codebase.

ATLAS calls SOUF AI before every LLM request. One standard. MIT. Four products. One mission: trustworthy AI.

OPEN SOURCE · MIT · REPRODUCIBLE · VERIFIABLE · UPSTREAM PR SUBMITTED

SOUF AI

SECURE OUTPUT FIREWALL

github.com/SRKRZ23/souf-ai

FULL EVIDENCE PACK

python scripts/run_all_benchmarks.py → 231 prompts · 5 benchmarks · ALL PASS
Ed25519 audit chain · Wilson 95% CI computed from 231 prompts (TP=188, TN=43)

REPRODUCE IN 2 COMMANDS:

```
git clone github.com/SRKRZ23/souf-ai && python scripts/run_all_benchmarks.py → Total: 231 prompts | ALL PASS
```

F1 SCORE
1.000

FALSE POS
0

FALSE NEG
0

P99 LATENCY
0.111ms

THROUGHPUT
17,553 req/s

Sardor Razikov · Independent ML Engineer · Top 1.9% Kaggle SPR · Zenodo Author

ORCID: 0009-0007-0731-4247 · GitHub: SRKRZ23 · X: @SardorRazi99093 · Tashkent, Uzbekistan

STAR THE REPO

github.com/SRKRZ23/souf-ai

MIT · HIPAA · PCI-DSS · Encoding attacks

UPSTREAM PR SUBMITTED

veeaio/lobstertrap · PR #open

SOUF AI patterns → Lobster Trap core

WILSON 95% CI

DENY: [0.980, 1.000]

ALLOW: [0.918, 1.000] · n=231

TechEx Intelligent Enterprise Solutions Hackathon · Track 1: Agent Security & AI Governance · May 2026

EU AI Act compliant · OWASP LLM01 · HIPAA · PCI-DSS · MIT License · Reproducible · Verifiable

THE QUESTION EVERY ENTERPRISE ASKS · WHY SOUF AI OVER LAKERA / NEMO / PROMPT GUARD

SOUF AI vs Commercial Guardrails

Feature parity matrix from public competitor docs · No fabricated competitor numbers

PROPERTY	SOUF AI v0.5f	Lakera Guard	NeMo Guardrails	Prompt Guard 2 (86M)
Latency P50	0.051 ms	~40–80 ms (SaaS)	~5–50 ms	~10–30 ms (BERT)
Throughput (1 core)	17,553 req/s	rate-limited	LLM-rail-bound	~100–300 req/s
Offline / air-gapped	✓ YES	✗ NO (SaaS)	✓ YES	✓ YES
Built-in HIPAA pack	✓ F1=1.000	guides only	custom Colang	✗
Built-in PCI-DSS pack	✓ F1=1.000	partial	custom Colang	✗
Audit chain (tamper-evident)	✓ Ed25519 + SHA-256	request logs	trace logs	✗
Wilson 95% CI reported	✓ YES	✗	✗	✗
License	MIT	Commercial closed	Apache 2.0	Llama 3 Community
Cost / 1M requests	\$0 self-hosted	per-seat license	\$0 + compute	\$0 + compute

SOUF AI is the only guardrail with all 6 wins.

Sub-ms latency · Offline-capable · Built-in HIPAA + PCI-DSS · Tamper-evident audit · Wilson CIs · MIT license · \$0 cost

Sources: Lakera Guard docs · NVIDIA NeMo Guardrails GitHub · Meta Prompt Guard model card · See docs/competitive_comparison.md